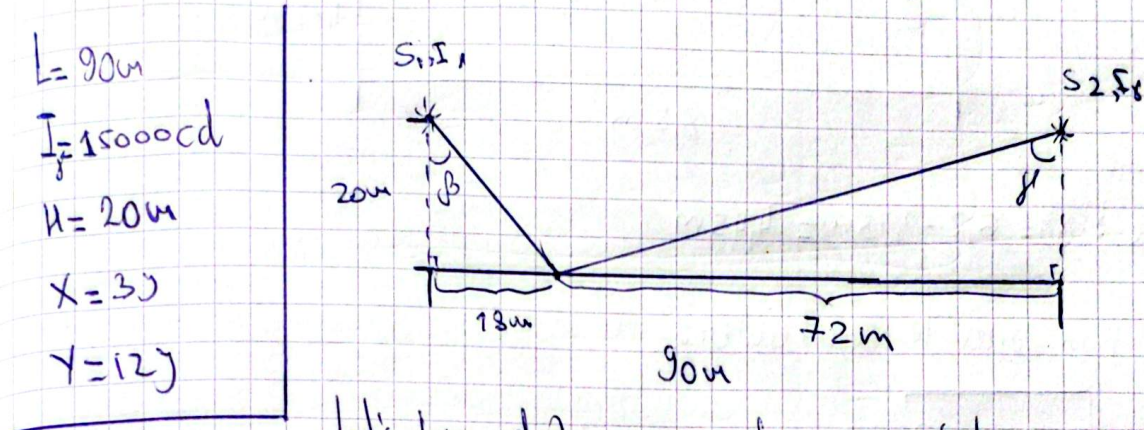


Problem 5:

2 uniformly radiating sources [90 m] apart, luminous intensity [15000 cd] located [20 m] height, is located at the intersection of the source line on the floor at 3/12 m distance from one of the sources.



Let's take distance from source 1 projection, for finding the point on the surface

$$X = 3y$$

$$Y = 12y$$

$$3y + 12y = 90$$

$$15y = 90$$

$$y = 6$$

$$3 \cdot 6 = 18\text{ m}$$

$$12 \cdot 6 = 72\text{ m}$$

} from Source 1 projection

We have to use cos law for calculating illuminance for each source:

$$E = \frac{I_1}{z^2} \cdot \cos \beta$$

$$E_1 = \frac{15000}{20^2} \cdot \frac{20}{\sqrt{20^2 + 18^2}} = \frac{15000}{20 \cdot 26,907} = 27,87 \text{ lx}$$

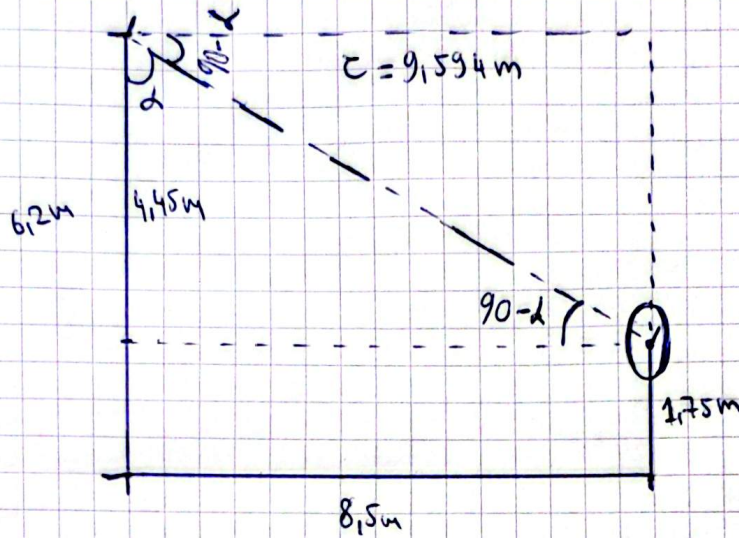
$$E_2 = \frac{15000}{20^2} \cdot \frac{20}{\sqrt{72^2 + 20^2}} = 10,03 \text{ lx}$$

$$[E_{\text{total}} = E_1 + E_2 = 37,9 \text{ lx total}]$$

[Problem 6:]

$$I_d = I_0 \cos(\alpha)$$

$$I_0 = 1100$$



Height difference: $\Delta H = 6.2 - 1.75 = 4.45\text{ m}$

Distance from source to sign center: $z = \sqrt{8.5^2 + 4.45^2} = 9.594\text{ m}$

$\cos \alpha = \frac{4.45}{9.594} = 0.464$ then $I_d = 1100 \cdot 0.464 = 510.4\text{ cd}$

Sign surface is vertical, for finding normal illuminance on its center:

$$E_n = \frac{I_d}{z^2} \cdot \cos(90 - \alpha) = \frac{I_d}{z^2} \cdot \sin \alpha =$$

$$\left[= \frac{510.4}{9.594^2} \cdot \frac{8.5}{9.594} = 4.91\text{ lx} \right]$$